

Notes: Hong, Lim & Stein (JF, 2000)

Bad News Travels Slowly: Size, Analyst Coverage, and the Profitability of Momentum Strategies

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	3
3.1	Sample and returns	3
3.2	Analyst coverage	4
3.3	Residual coverage	4
3.4	Momentum portfolios	4
4	Models and empirical specifications	4
4.1	Momentum by size	4
4.2	Coverage residual sorts	5
4.3	Bad-news asymmetry	5
4.4	Robustness regressions	5
5	Identification and empirical design	5
5.1	What is identified	5
5.2	Why size and coverage must be separated	5
5.3	Why losers are the sharpest test	6
5.4	Main threats and how the paper responds	6
6	Tables and figures	7
6.1	Table I: Descriptive Statistics for Analyst Coverage; Table II: Determinants of Analyst Coverage	7
6.2	Table III: Momentum Strategies Using Raw Returns and Sorting by Size; Figure 1: Momentum Profits and Firm Size	7
6.3	Table IV: Momentum Strategies Using Raw Returns and Sorting by Model 1 Residuals; Table V: Momentum Strategies Using Raw Returns and Sorting by Size and Model 1 Residuals . .	8
6.4	Table VI: Momentum Strategies Using Beta-Adjusted Returns; Table VII: Turnover-Adjusted Residuals	9
6.5	Table VIII: Momentum Strategies for Subperiods; Figure 2: Cumulative Beta-Adjusted Returns in Event Time	9
6.6	Table IX: Cross-Sectional Momentum Regressions, 1979–1992	10
7	Short summary	11
8	Conclusion	12
8.1	Core conclusion	12
8.2	Economic intuition	12

8.3	Incremental contribution	12
8.4	Limitations	12
8.5	Takeaway	12

1 Big picture

- **Question.** Why do medium-horizon momentum strategies work, and is momentum stronger where firm-specific information diffuses more slowly?
- **Theory tested.** The paper tests the gradual-information-diffusion model of Hong and Stein (1999): prices underreact when value-relevant information spreads slowly through the investing public.
- **Main proxy for slow diffusion.** Analyst coverage. Stocks with fewer analysts should have slower information diffusion, especially when the firm is small and information is costly to collect.
- **Main findings.** Momentum profits decline with size once the very smallest firms are excluded; conditional on size, momentum is stronger for low-analyst-coverage stocks; and analyst coverage matters more for losers than winners.
- **Bad-news asymmetry.** The strongest evidence comes from past losers. Negative information appears to diffuse more slowly than positive information, so loser portfolios display stronger continuation when coverage is low.
- **Economic takeaway.** Momentum is not just a generic return pattern. Its profitability is linked to the information environment of the stock: smaller and less-followed firms display stronger underreaction.

2 Incremental contribution of the paper

- **Direct test of a specific momentum theory.** Earlier work documents momentum. This paper tests the cross-sectional prediction of the gradual-information-diffusion theory: momentum should be stronger when information diffusion is slower.
- **Analyst coverage as information-diffusion proxy.** The paper uses residual analyst coverage, purged of size and other covariates, as a proxy for investor attention and information flow.
- **Separates size from coverage.** Size and analyst coverage are highly correlated. The paper creates coverage residuals and two-way sorts to show that both variables contain information, but analyst coverage has distinct explanatory power.
- **Bad-news channel.** The paper documents that coverage effects are stronger for past losers than for past winners, linking momentum to delayed incorporation of negative information.
- **Multiple robustness checks.** The results survive beta-adjusted returns, turnover controls, industry controls, options-listing controls, subperiod splits, event-time cumulative returns, and Fama-MacBeth momentum regressions.
- **Bridge between behavioral finance and information economics.** The paper moves beyond documenting return predictability and shows how limits to information dissemination can shape asset-pricing anomalies.

3 Data and measurement

3.1 Sample and returns

The sample consists of NYSE, AMEX, and Nasdaq stocks. Momentum portfolios are formed using six-month lagged returns and held for six months. The core postformation outcome is the average monthly return of past winners minus past losers.

Interpretation: The sample design follows the standard Jegadeesh–Titman momentum design, but the paper modifies the analysis to study how momentum varies with firm size and analyst coverage.

3.2 Analyst coverage

Analyst coverage is measured using I/B/E/S: the number of analysts providing fiscal-year-one earnings estimates. If a CRSP firm cannot be matched to I/B/E/S, coverage is set to zero. Coverage is often measured six months before the return ranking window begins.

Interpretation: Using lagged coverage reduces the concern that analysts respond to the same price moves that define winner and loser portfolios. It also makes coverage closer to a pre-existing information-environment variable.

3.3 Residual coverage

The main coverage measure is residual coverage from monthly cross-sectional regressions such as:

$$\log(1 + \text{Analysts}_i) = \alpha + \beta_1 \log(\text{Size}_i) + \beta_2 \text{Nasdaq}_i + u_i.$$

Stocks are sorted into low, medium, and high residual coverage classes. The least-covered firms are Sub1 and the most-covered firms are Sub3.

Interpretation: Residual coverage isolates coverage variation not mechanically explained by firm size. This is crucial because size is the dominant predictor of analyst coverage.

3.4 Momentum portfolios

Past-return portfolios are usually:

- P1: bottom 30 percent of lagged six-month returns;
- P2: middle 40 percent;
- P3: top 30 percent.

The baseline momentum measure is:

$$P3 - P1.$$

The paper also studies a “LAST” strategy that focuses on the bad-news asymmetry:

$$LAST = P1/Sub3 - P1/Sub1.$$

Interpretation: P3-P1 is standard momentum. LAST asks whether low-coverage losers continue to fall more than high-coverage losers, which is the sharpest test of slow bad-news diffusion.

4 Models and empirical specifications

4.1 Momentum by size

The first empirical specification is a double-sort by lagged return and size:

$$\text{Return}_{post} = f(\text{PastReturnClass}, \text{SizeClass}).$$

Interpretation: If momentum is stronger where information diffuses slowly, it should be stronger for smaller stocks after excluding the tiniest stocks with measurement and microstructure problems.

4.2 Coverage residual sorts

The baseline coverage analysis sorts stocks by past return and residual analyst coverage:

$$Return_{post} = f(PastReturnClass, ResidualCoverageClass).$$

The main comparison is:

$$(P3 - P1)_{Sub1} - (P3 - P1)_{Sub3}.$$

Interpretation: A positive difference means momentum is more profitable among low-coverage stocks than high-coverage stocks, conditional on firm size and Nasdaq status.

4.3 Bad-news asymmetry

The paper decomposes the P3-P1 effect into winners and losers. The key loser-side comparison is:

$$LAST = P1/Sub3 - P1/Sub1.$$

Interpretation: If bad news travels slowly, low-coverage losers should continue to underperform high-coverage losers. This is why the paper emphasizes the LAST strategy.

4.4 Robustness regressions

The Fama-MacBeth regression uses a serial-correlation coefficient as the dependent variable:

$$RHO_i = \alpha + \beta_1 \log(1 + Coverage_i) + \beta_2 \log(Size_i) + \beta_3 Nasdaq_i + \varepsilon_i.$$

The interactive version adds a coverage-by-size term.

Interpretation: A negative coefficient on coverage means that more analyst coverage lowers medium-horizon return continuation. The interaction asks whether coverage matters less as firms become larger.

5 Identification and empirical design

5.1 What is identified

The paper does not estimate a causal effect of assigning more analysts to a firm. Instead, it tests whether the cross-sectional pattern of momentum is consistent with gradual information diffusion.

Interpretation: The design is theory-guided predictive evidence. Analyst coverage is a proxy for the speed of information diffusion; the test is whether momentum is stronger where the proxy predicts slower diffusion.

5.2 Why size and coverage must be separated

Firm size strongly predicts analyst coverage. A naive coverage sort could simply replicate a size sort. The paper therefore uses residual coverage after controlling for log size and a Nasdaq dummy, and then repeats the analysis within size classes.

Interpretation: The two-way sorts in Table V are central because they make the coverage comparison within narrower size bins. This reduces the risk that coverage is only standing in for size.

5.3 Why losers are the sharpest test

The gradual-information-diffusion story is especially natural for bad news because short-sale constraints, lack of attention, and reluctance to process negative information can slow price adjustment. The paper predicts and finds that the coverage effect is stronger for losers than winners.

Interpretation: This is the cleanest mechanism test. If coverage merely proxied for liquidity or risk, the asymmetry between winners and losers would be harder to explain.

5.4 Main threats and how the paper responds

- **Size confounding:** handled using residual coverage and size disaggregation.
- **Risk-factor confounding:** handled using beta-adjusted returns and market-model adjustments.
- **Turnover/liquidity confounding:** handled by residuals from a model including turnover.
- **Industry composition:** checked with industry controls in the coverage model.
- **Subperiod instability:** checked over 1980–1984, 1985–1990, and 1991–1996.
- **Portfolio-sorting artifact:** checked through Fama-MacBeth cross-sectional momentum regressions.

Interpretation: The evidence is strongest because multiple designs point in the same direction: momentum is stronger in smaller, low-coverage stocks, especially among losers.

6 Tables and figures

6.1 Table I: Descriptive Statistics for Analyst Coverage; Table II: Determinants of Analyst Coverage

Read: Table I shows that analyst coverage rises sharply over time but remains extremely sparse for small firms. Table II shows that firm size is by far the dominant determinant of coverage, although beta, returns, value, Nasdaq, turnover, and options listing also matter.

Economic message: The paper's central proxy, analyst coverage, is strongly linked to firm size. Therefore coverage must be residualized before it can be used as an information-diffusion proxy.

Interpretation: Table I motivates dropping the very smallest firms from coverage tests because there is almost no variation in coverage. Table II motivates the residual-coverage procedure.

Table I Descriptive Statistics for Analyst Coverage													
Descriptive statistics for analyst coverage for NYSE, AMEX, and Nasdaq stocks, excluding ADRs, REITs, closed-end funds, and primes and scores the period 1976 to 1996. Panel A reports for the even years between 1976 and 1996 the number of firms in the sample, their mean and size, the number of analysts at various coverage percentiles, and the percentage of firms that had no coverage. Panel B reports for 1988 the same statistics as in Panel A.													
Panel A: All Stocks, 1976-1996													
No. of Firms	Mean Size (millions)	Median Size (millions)	No. of Analysts at Coverage Percentiles										Percentage of firms uncovered
			10	20	30	40	50	60	70	80	90		
4402	183.6	18.7	0	0	0	0	0	0	0	1	4	77.3%	
4472	176.4	22.7	0	0	0	0	0	0	0	2	5	71.5%	
4329	245.9	34.6	0	0	0	0	0	1	2	4	9	58.2%	
4754	249.3	30.3	0	0	0	0	0	1	2	5	11	59.3%	
5049	332.3	44.4	0	0	0	0	0	1	3	6	12	50.8%	
5364	387.4	42.5	0	0	0	0	0	1	3	6	14	50.5%	
5932	402.2	32.6	0	0	0	0	0	1	3	5	12	50.1%	
5567	500.7	34.5	0	0	0	0	1	2	3	7	13	45.4%	
5438	672.8	49.8	0	0	0	0	1	2	3	6	13	46.7%	
5890	802.9	81.1	0	0	0	0	1	3	4	7	13	40.0%	
6460	978.1	90.8	0	0	0	1	2	3	4	7	12	36.9%	

Panel B: Breakdown of Analyst Coverage by Firm Size for 1988														
NYSE/AMEX Breakpoints	No. of Firms	Mean Size (millions)	Median Size (millions)	No. of Analysts at Coverage Percentiles										Percentage of firms uncovered
				10	20	30	40	50	60	70	80	90		
the 20th percentile	2597	9.6	8.3	0	0	0	0	0	0	0	0	1	82.0%	
in the 20th & 40th percentiles	1363	45.1	42.5	0	0	0	0	1	1	2	3	4	41.7%	
in the 40th & 60th percentiles	937	147.1	133.3	0	0	1	2	3	4	5	7	9	21.5%	
in the 60th & 80th percentiles	607	554.0	495.8	1	4	6	7	8	10	12	14	17	7.7%	
the 80th percentile	431	4235.7	2390.7	8	13	16	19	21	23	26	28	30	5.6%	

(a) Table I: analyst coverage distribution over time and by firm size.

Table II
Determinants of Analyst Coverage, 12/1988

Dependent variable is $\log(1 + \text{Analyst coverage})$. Log Size is the log of a firm's year-end market value. NASD is a Nasdaq dummy. Book/Mkt is the ratio of a firm's year-end book-to-market value. Beta is a firm's market beta. P is a firm's share price. Var is the variance of a firm's return the last 200 observations from year-end. R_k is the rate of return of a firm lagged k years for $k = 0, 1, 2, 3, 4$. T-O is a firm's turnover defined prior six months' trading volume divided by shares outstanding. NASD * T-O is the Nasdaq dummy times firm turnover. OPT is a dummy whether a firm has options trading on CBOE, NYSE, AMEX, Philadelphia, or Pacific stock exchanges. IND is a set of CRSP industry dummies. There are 2,012 observations. t -statistics are in parentheses.

Log Size	NASD	Book/Mkt	Beta	1/P	Var	R_0	R_1	R_2	R_3	R_4	T-O	NASD * T-O	OPT	IND	R^2	
0.54	0.03														No	0.61
(32.67)	(0.99)														Yes	0.61
0.56	0.04														No	0.61
(32.90)	(1.21)														Yes	0.61
0.55	0.05	0.12													No	0.61
(33.03)	(1.50)	(3.15)													Yes	0.61
0.57	0.07	0.17													No	0.61
(32.22)	(2.00)	(4.30)													Yes	0.61
0.50	0.07		0.38												Yes	0.61
(48.41)	(2.28)		(11.54)												Yes	0.61
0.51	0.09	0.40													Yes	0.61
(46.11)	(2.62)	(10.94)													Yes	0.61
0.57	0.09			-0.52	-1.27	-0.50	-0.28	-0.28	-0.04	-0.16					Yes	0.62
(49.87)	(2.59)			(-3.12)	(-3.23)	(-9.46)	(-6.06)	(-6.00)	(-0.85)	(-3.46)					Yes	0.62
0.52	-0.02										3.82	-0.53			No	0.64
(33.46)	(-0.54)										(8.18)	(-0.83)			Yes	0.64
0.50	-0.02										3.52	-0.37	0.12		No	0.64
(38.83)	(-0.48)										(7.32)	(-0.64)	(2.48)		Yes	0.64

(b) Table II: cross-sectional determinants of analyst coverage.

6.2 Table III: Momentum Strategies Using Raw Returns and Sorting by Size; Figure 1: Momentum Profits and Firm Size

Read: Table III and Figure 1 show an inverted-U relation between firm size and momentum profits. Momentum is negative for the smallest stocks, strongest in the second and third size deciles, and declines among larger firms.

Economic message: The size pattern supports gradual information diffusion after excluding tiny stocks with microstructure and measurement issues. Medium-small firms have enough liquidity to avoid extreme reversal but still enough information friction to display strong momentum.

Interpretation: Figure 1 is the cleanest visual summary of the size result. It shows why the paper focuses on stocks above the 20th NYSE/AMEX percentile.

Table III
Momentum Strategies, 1/1980-12/1996, Using Raw Returns and Sorting by Size
ble includes all stocks. The relative momentum portfolios are formed based on six-month lagged raw returns and held for six months. The are ranked in ascending order on the basis of six-month lagged returns. Portfolio P1 is an equally weighted portfolio of stocks in the performing 30 percent, portfolio P2 includes the middle 40 percent, and portfolio P3 includes the best-performing 30 percent. This table the average monthly returns of these portfolios and portfolios formed using size-based subsamples of stocks. Using NYSE/AMEX decile ints, the smallest firms are in size class 1, the next in 2, and the largest are in 10. Mean (median) size is in millions. *t*-statistics are in reses.

ast	All Stocks	Size Class (NYSE/AMEX Decile Breakpoints)									
		1	2	3	4	5	6	7	8	9	10
1	0.01043 (2.44)	0.02106 (4.44)	0.00653 (1.37)	0.00231 (0.52)	0.00194 (0.43)	0.00469 (1.05)	0.00573 (1.32)	0.00606 (1.43)	0.01010 (2.51)	0.00922 (2.25)	0.01258 (3.37)
	0.01378 (4.48)	0.01662 (4.97)	0.01290 (3.84)	0.01280 (3.88)	0.01244 (3.75)	0.01395 (4.18)	0.01374 (4.14)	0.01375 (4.27)	0.01393 (4.40)	0.01401 (4.43)	0.01355 (4.50)
	0.01570 (4.35)	0.01733 (4.40)	0.01507 (3.89)	0.01664 (4.35)	0.01570 (4.05)	0.01655 (4.26)	0.01608 (4.26)	0.01491 (4.13)	0.01436 (4.04)	0.01363 (3.96)	0.01278 (3.84)
	0.00527 (2.61)	-0.00374 (-1.77)	0.00854 (3.60)	0.01433 (6.66)	0.01376 (6.10)	0.01187 (5.32)	0.01035 (4.80)	0.00885 (3.72)	0.00425 (1.90)	0.00441 (1.73)	0.00021 (0.08)
	—	—	0.746	0.732	0.763	0.780	0.774	0.869	0.901	1.086	—
	size	7	21	44	79	138	242	437	806	1658	7290
	analyst	0.1	0.5	1.1	2.0	3.2	5.0	7.3	10.6	15.3	21.4
	analyst	0.0	0.0	0.7	1.3	2.5	4.4	6.9	10.5	15.7	22.4
	—	—	—	—	—	—	—	—	—	—	—
	—	—	—	—	—	—	—	—	—	—	—

(a) Table III: P1/P2/P3 momentum returns by size decile.

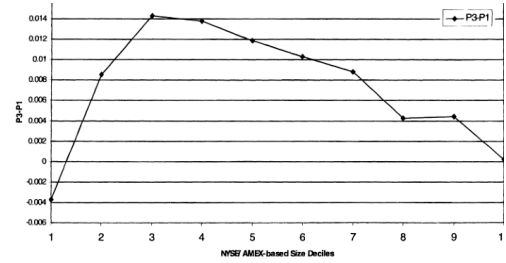


Figure 1. Momentum profits and firm size. Momentum profits (P3 - P1) plotted again NYSE/AMEX-based size deciles, 1 (smallest) to 10 (largest).

(b) Figure 1: momentum profits by firm-size decile.

6.3 Table IV: Momentum Strategies Using Raw Returns and Sorting by Model 1 Residuals; Table V: Momentum Strategies Using Raw Returns and Sorting by Size and Model 1 Residuals

Read: Table IV shows stronger momentum for low-coverage stocks than high-coverage stocks after excluding the smallest firms. Table V repeats the analysis within size classes and shows the coverage effect is strongest in smaller size classes.

Economic message: Analyst coverage predicts where momentum is strongest. The two-way sort makes the comparison credible because the low- and high-coverage groups are better matched on size.

Interpretation: The most important evidence is the loser side: low-coverage losers continue to perform worse than high-coverage losers. This is consistent with slow diffusion of bad firm-specific news.

Table IV
Momentum Strategies, 1/1980-12/1996, Using Raw Returns and Sorting by Model 1 Residuals

This table includes only stocks above the NYSE/AMEX 20th percentile. The relative momentum portfolios are formed based on six-month lagged raw returns and held for six months. The stocks are ranked in ascending order on the basis of six-month lagged returns. Portfolio P1 is an equally weighted portfolio of stocks in the worst-performing 30 percent, portfolio P2 includes the middle 40 percent, and portfolio P3 includes the best-performing 30 percent. This table reports the average monthly returns of these portfolios and portfolios formed using an independent sort on Model 1 analyst coverage residuals of log size and a Nasdaq dummy. The least-covered firms are in Sub1, the medium covered firms in Sub2, the most covered firms in Sub3. Mean (median) size is in millions. *t*-statistics are in parentheses.

Past	All Stocks	Residual Coverage Class			
		Low: Sub1	Medium: Sub2	High: Sub3	Sub1 - Sub3
P1	0.00622 (1.54)	0.00271 (0.66)	0.00669 (1.70)	0.00974 (2.31)	-0.00703 (-5.16)
P2	0.01367 (4.40)	0.01257 (4.20)	0.01397 (4.58)	0.01439 (4.29)	-0.00182 (-2.11)
P3	0.01562 (4.35)	0.01402 (3.95)	0.01583 (4.52)	0.01690 (4.45)	-0.00288 (-2.80)
P3 - P1	0.00940 (4.89)	0.01131 (5.46)	0.00915 (4.64)	0.00716 (3.74)	0.00415 (3.50)
Mean size	962	962	986	455	
Median size	103	103	200	180	
Mean analyst	1.5	1.5	6.7	9.7	
Median analyst	0.1	0.1	3.5	7.6	

(a) Table IV: raw-return momentum by residual coverage.

Table V
Momentum Strategies, 1/1980-12/1996, Using Raw Returns and Sorting by Size and Model 1 Residuals
e includes only stocks above the NYSE/AMEX 20th percentile. The relative momentum portfolios are formed based on six-month lagged ns and held for six months. The stocks are ranked in ascending order on the basis of six-month lagged returns. Portfolio P1 is an equally portfolio of stocks in the worst-performing 30 percent, portfolio P2 includes the middle 40 percent, and portfolio P3 includes the orning 30 percent. This table reports the average monthly returns to portfolios formed by sorts on size and Model 1 analyst coverage of log size and a Nasdaq dummy. Size is sorted using NYSE/AMEX breakpoints. The least covered firms are in Sub1, the medium firms in Sub2, the most covered firms in Sub3. Mean (median) size is in millions. *t*-statistics are in parentheses.

Coverage Class	Size Class:			
	1: 20th-40th Percentile	2: 40th-60th Percentile	3: 60th-80th Percentile	4: 80th-100th Percentile
P3 - P1 = 0.01511 (6.46)	P3 - P1 = 0.01057 (4.49)	P3 - P1 = 0.00605 (3.11)	P3 - P1 = 0.00092 (0.49)	
size	63	199	653	5056
size	59	183	592	2263
coverage	0.0	0.6	3.7	11.1
Sub2	P3 - P1 = 0.01389 (5.48)	P3 - P1 = 0.00975 (4.95)	P3 - P1 = 0.00316 (1.62)	P3 - P1 = 0.00009 (0.05)
size	61	207	678	5163
size	56	193	629	2853
coverage	0.9	3.6	9.0	18.8
Sub3	P3 - P1 = 0.01147 (5.10)	P3 - P1 = 0.00730 (3.60)	P3 - P1 = 0.00424 (2.02)	P3 - P1 = 0.00070 (0.33)
size	64	202	663	3650
size	61	188	615	2511
coverage	3.1	7.6	14.7	24.9
Sub3	P3 - P1 = 0.00364 (2.13)	P3 - P1 = 0.00327 (1.95)	P3 - P1 = 0.00180 (1.18)	P3 - P1 = 0.00023 (.14)

(b) Table V: raw-return momentum by size and residual coverage.

6.4 Table VI: Momentum Strategies Using Beta-Adjusted Returns; Table VII: Turnover-Adjusted Residuals

Read: Table VI shows that the main effects survive beta adjustment. Table VII shows that using residual coverage from a model including turnover still produces a large coverage differential.

Economic message: The results are not simply compensation for beta or a turnover/liquidity artifact. The coverage-based information-diffusion interpretation survives these risk and trading-friction controls.

Interpretation: The robustness checks strengthen the claim that analyst coverage measures information diffusion rather than only risk, liquidity, or transactions costs.

Table VI
Momentum Strategies, 1/1980–12/1996, Using Beta-Adjusted Returns and Sorting by Model 1 Residuals

This table includes only stocks above the NYSE/AMEX 20th percentile. The relative momentum portfolios are formed based on six-month lagged beta-adjusted returns and held for six months. The stocks are ranked in ascending order on the basis of six-month lagged returns. Portfolio P1 is an equally weighted portfolio of stocks in the worst-performing 30 percent, portfolio P2 includes the middle 40 percent, and portfolio P3 includes the best-performing 30 percent. This table reports the average monthly beta-adjusted returns of these portfolios and portfolios formed using an independent sort on Model 1 analyst coverage residuals of log size and a Nasdaq dummy. The least covered firms are in Sub1, the medium covered firms in Sub2, the most covered firms in Sub3. Mean (median) size is in millions. *t*-statistics are in parentheses.

Past	All Stocks	Residual Coverage Class			
		Low: Sub1	Medium: Sub2	High: Sub3	Sub1 – Sub3
P1	–0.00753 (–3.29)	–0.01007 (–3.97)	–0.00712 (–3.30)	–0.00511 (–2.13)	–0.00497 (–3.64)
P2	0.00280 (2.44)	0.00313 (2.48)	0.00299 (2.92)	0.00231 (1.73)	0.00081 (1.06)
P3	0.00444 (3.17)	0.00423 (2.76)	0.00454 (3.50)	0.00430 (2.74)	–0.00006 (–0.06)
P3 – P1	0.01197 (5.99)	0.01431 (6.79)	0.01167 (5.76)	0.00940 (4.62)	0.00491 (4.04)
Mean size		1070	998	464	
Median size		106	221	186	
Mean analyst		1.8	7.1	9.9	
Median analyst		0.2	4.0	7.9	

(a) Table VI: beta-adjusted momentum by residual coverage.

Table VII
Momentum Strategies, 1/1984–12/1996, Using Raw Returns and Sorting by Model 8 Residuals

This table includes only stocks above the NYSE/AMEX 20th percentile. The relative momentum portfolios are formed based on six-month lagged raw returns and held for six months. The stocks are ranked in ascending order on the basis of six-month lagged returns. Portfolio P1 is an equally weighted portfolio of stocks in the worst-performing 30 percent, portfolio P2 includes the middle 40 percent, and portfolio P3 includes the best-performing 30 percent. This table reports the average monthly returns of these portfolios and portfolios formed using an independent sort on Model 8 analyst coverage residuals of log size, a Nasdaq dummy, firm turnover, and Nasdaq dummy times firm turnover. The least covered firms are in Sub1, the medium covered firms in Sub2, the most covered firms in Sub3. Mean (median) size is in millions. *t*-statistics are in parentheses.

Past	All Stocks	Residual Coverage Class			
		Low: Sub1	Medium: Sub2	High: Sub3	Sub1 – Sub3
P1	0.00498 (1.11)	0.00190 (0.42)	0.00553 (1.26)	0.00747 (1.56)	–0.00557 (–3.58)
P2	0.01209 (3.44)	0.01126 (3.44)	0.01273 (3.67)	0.01229 (3.16)	–0.00103 (–1.00)
P3	0.01351 (3.38)	0.01210 (3.20)	0.01377 (3.54)	0.01458 (3.31)	–0.00248 (–2.11)
P3 – P1	0.00853 (4.22)	0.01020 (4.67)	0.00824 (3.92)	0.00711 (3.46)	0.00309 (2.23)
Mean size		1412	1078	442	
Median size		124	282	180	
Mean analyst		2.9	8.3	10.0	
Median analyst		0.4	4.9	7.7	

(b) Table VII: momentum with turnover-adjusted coverage residuals.

6.5 Table VIII: Momentum Strategies for Subperiods; Figure 2: Cumulative Beta-Adjusted Returns in Event Time

Read: Table VIII shows that the coverage effect appears in all three subperiods. Figure 2 shows that low-coverage momentum profits cumulate for many months, while high-coverage profits are smaller and converge faster.

Economic message: The pattern is persistent over time and dynamic. Low-coverage stocks do not only underreact at the six-month horizon; they continue adjusting in event time.

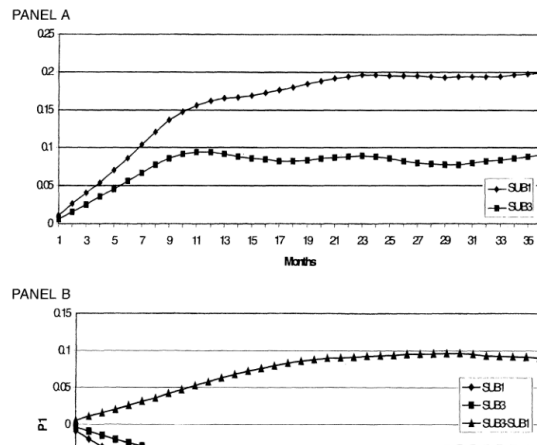
Interpretation: Panel B of Figure 2 is especially important because it plots the LAST strategy and shows that loser-side coverage differences cumulate over time.

Table VIII
Momentum Strategies for Subperiods, 1/1980–12/1996,
Using Raw Returns and Sorting by Model 1 Residuals

This table includes only stocks above the NYSE/AMEX 20th percentile. The relative momentum portfolios are formed based on six-month lagged raw returns and held for six months. The stocks are ranked in ascending order on the basis of six-month lagged returns. Portfolio P1 is an equally weighted portfolio of stocks in the worst-performing 30 percent and portfolio P3 includes the best-performing 30 percent. This table reports the average monthly returns of these portfolios and portfolios formed using an independent sort on Model 1 analyst coverage residuals of log size and a Nasdaq dummy. The least covered firms are in Sub1, the medium covered firms in Sub2, the most covered firms in Sub3. Mean (median) size is in millions. *t*-statistics are in parentheses.

Subperiods	Past	All Stocks	Residual Coverage Class			
			Low: Sub1	Medium: Sub2	High: Sub3	Sub1 – Sub3
1/1980–12/1984	P1	0.00713 (0.94)	0.00282 (0.35)	0.00806 (1.09)	0.01215 (1.63)	–0.00933 (–3.48)
	P3	0.01852 (2.62)	0.01706 (2.30)	0.01850 (2.69)	0.01991 (2.79)	–0.00286 (–1.31)
	P3 – P1	0.01139 (3.27)	0.01424 (3.61)	0.01044 (2.88)	0.00777 (2.52)	0.00647 (2.90)
1/1985–12/1990	P1	–0.00302 (–0.41)	–0.00617 (–0.85)	–0.00205 (–0.28)	–0.00081 (–0.10)	–0.00536 (–2.21)
	P3	0.01079 (1.54)	0.00920 (1.38)	0.01164 (1.70)	0.01145 (1.50)	–0.00225 (–1.29)
	P3 – P1	0.01381 (4.65)	0.01538 (4.86)	0.01369 (4.42)	0.01227 (4.05)	0.00311 (1.62)
1/1991–12/1996	P1	0.01472 (2.49)	0.01151 (1.91)	0.01428 (2.49)	0.01828 (2.95)	–0.00677 (–3.34)

(a) Table VIII: subperiod robustness for residual-coverage momentum.



(b) Figure 2: event-time cumulative beta-adjusted returns.

6.6 Table IX: Cross-Sectional Momentum Regressions, 1979–1992

Read: Table IX uses Fama-MacBeth cross-sectional regressions to show that analyst coverage lowers six-month return autocorrelation. The interaction specification shows that the effect of coverage declines with firm size.

Economic message: This regression approach confirms the portfolio results without relying on fixed portfolio cutoffs. Coverage reduces momentum, and its role is strongest where information frictions are likely largest.

Interpretation: Table IX is the final robustness check. It converts the sort-based evidence into a continuous regression framework and shows the same economic message.

Table IX
Cross-Sectional Momentum Regressions, 1979–1992

This table includes only stocks above the NYSE/AMEX 20th percentile. The dependent variable is RHO: regression coefficient of six-month returns (net risk-free) on lagged six-month returns. Panel A: Independent variables are $\log(1 + \text{Analyst coverage})$, log size, and a Nasdaq dummy. Panel B: Independent variables are $\log(1 + \text{Analyst coverage})$, log size, interaction of $\log(1 + \text{Analyst coverage})$ and log size and a Nasdaq dummy. Note: t -statistics are adjusted for serial correlation.

Panel A						
Year	Coverage	t -statistics	Size	t -statistics		
79	−0.0015	−0.1800	−0.0097	−1.7530		
80	−0.0014	−0.2040	−0.0188	−3.8600		
81	−0.0039	−0.6090	−0.0061	−1.2800		
82	0.0040	0.5500	−0.0259	−4.5520		
83	−0.0136	−1.9020	0.0050	0.8990		
84	−0.0280	−3.9300	0.0168	2.9200		
85	−0.0166	−2.1330	0.0146	2.4060		
86	−0.0357	−5.6310	0.0240	4.7650		
87	−0.0111	−1.8160	0.0040	0.8480		
88	−0.0163	−2.5820	−0.0108	−2.2560		
89	−0.0141	−2.2900	−0.0071	−1.5550		
90	−0.0208	−3.2060	−0.0004	−0.0860		
91	−0.0126	−1.7100	0.0059	1.1680		
92	−0.0031	−0.4720	0.0019	0.4070		
Fama–MacBeth	−0.0125	−3.8023	−0.0005	−0.1265		
Pooled with year dummies	−0.0127	−5.0898	−0.0004	−0.2832		

Panel B						
Year	Coverage	t -statistics	Size	t -statistics	Interaction: Coverage * Size	t -statistics
79	0.0306	0.7260	−0.0060	−0.8320	−0.0027	−0.775
80	0.1000	2.5930	−0.0064	−0.9480	−0.0084	−2.674
81	0.0055	0.1430	−0.0049	−0.6980	−0.0008	−0.248
82	−0.0382	−0.8500	−0.0321	−3.7080	0.0035	0.951
83	−0.0053	−0.1270	0.0061	0.7730	−0.0007	−0.205
84	−0.1441	−3.3310	−0.0001	−0.0060	0.0095	2.721
85	−0.1618	−3.3860	−0.0092	−0.9330	0.0118	3.079
86	−0.0457	−1.1950	0.0224	2.8280	0.0008	0.265
87	−0.0664	−1.8020	−0.0051	−0.6720	0.0044	1.521
88	−0.1622	−4.2580	−0.0359	−4.4700	0.0118	3.884
89	−0.0837	−2.4370	−0.0189	−2.5800	0.0057	2.059
90	−0.1372	−3.6960	−0.0212	−2.6360	0.0094	3.184
91	−0.0898	−2.2350	−0.0084	−0.9450	0.0063	1.954
92	−0.0836	−2.3560	−0.0118	−1.5720	0.0065	2.308
Fama–MacBeth	−0.0630	−1.8920	−0.0094	−2.3701	0.0041	1.5423

Table IX: Fama-MacBeth cross-sectional momentum regressions.

7 Short summary

- The paper tests whether momentum profits are stronger where information diffuses slowly.
- Analyst coverage is used as a proxy for information diffusion.
- Size is the strongest determinant of coverage, so the paper residualizes coverage and performs two-way size-coverage sorts.
- Momentum is strongest outside the tiniest stocks and declines with size.
- Within size groups, momentum is stronger for low-coverage stocks.
- The effect is especially strong for past losers, supporting the idea that bad news travels slowly.
- The results survive beta adjustment, turnover controls, industry controls, subperiod splits, and Fama-MacBeth regressions.

8 Conclusion

8.1 Core conclusion

Hong, Lim, and Stein show that momentum is linked to the speed of information diffusion. Stocks with less analyst coverage and smaller size exhibit stronger continuation, especially after negative past returns.

8.2 Economic intuition

When fewer analysts follow a firm, firm-specific news reaches investors more slowly. Prices therefore underreact initially and continue to adjust over subsequent months. Bad news is especially slow to travel, generating stronger continuation among losers.

8.3 Incremental contribution

The paper turns the general momentum anomaly into a test of a specific theory. Its contribution is not merely documenting another cross-sectional predictor of momentum; it shows that a plausible measure of information flow predicts where momentum should be strongest.

8.4 Limitations

Analyst coverage is a proxy, not a randomized treatment. Coverage may also capture liquidity, institutional ownership, short-sale constraints, or firm visibility. The paper's many controls reduce these concerns but do not fully eliminate them.

8.5 Takeaway

Momentum is partly an information-diffusion phenomenon. Bad news travels slowly, and it travels especially slowly for small, neglected stocks.